

Sport injuries and joint reconstruction

MSK BLOCK 2

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what are sport injuries

refers to the kind of injuries that most commonly occur during sports or exercise.

- Some sports injuries result from accidents: others due to poor training practices, improper equipment, lack of conditioning, or insufficient warm-up or stretching.

Acute injuries

- ▶ Characterized by sudden appearance of symptoms usually associated with a single traumatic incident.
 - ⇒ Strains :
 - ⇒ Sprains
 - ⇒ Joint dislocation
 - ⇒ Contusions
 - ⇒ Bone fractures

Chronic injuries

- ▶ Also called overuse injuries or stress injuries.
 - Gradual onset and are caused by repetitive light trauma to soft tissue or bone.
 - Includes :
 - ⇒ *Tennis elbow*
 - ⇒ Shin splints
 - ⇒ Bursitis.
 - ⇒ Stress fractures
- All injuries occur due to:
Direct Blows, Torsions or Shearing
Soft Tissue – skin, organs, tendons, ligaments
Hard Tissues - bone

Common sport injuries

- ▶ Ankle sprains
 - Quads sprain
 - Hamstring strain
 - Shin splints
 - ACL injury
 - Golfer's elbow
 - Tennis elbow
 - Planter fasciatis
 - LBP
 - Runner's knee

Acute injury management

- ▶ R – Rest – 2-3 days with immobilization
 - I – Ice – Reduces pain and spasm, minimizes cell death and causes vasoconstriction
 - C – Compression – Decreases swelling by slowing the flow of fluid to the area
 - E – Elevation – Decreases swelling by encouraging blood to return to the heart

Chronic injury management

- ▶ Pain killers
- ▶ • Immobilization
- ▶ • Corticosteroid injections
- ▶ • Physiotherapy
- ▶ • Surgery
- ▶ • Rehabilitation

Rehabilitation

- ▶ Start with alternative training
 - ▶ • Circuit training
 - ▶ • Conditioning
 - ▶ • Endurance
 - ▶ • Flexibility
 - ▶ • Strength training
 - ▶ • Stretching techniques
 - ▶ • Warm-up
 - ▶ • Weight training

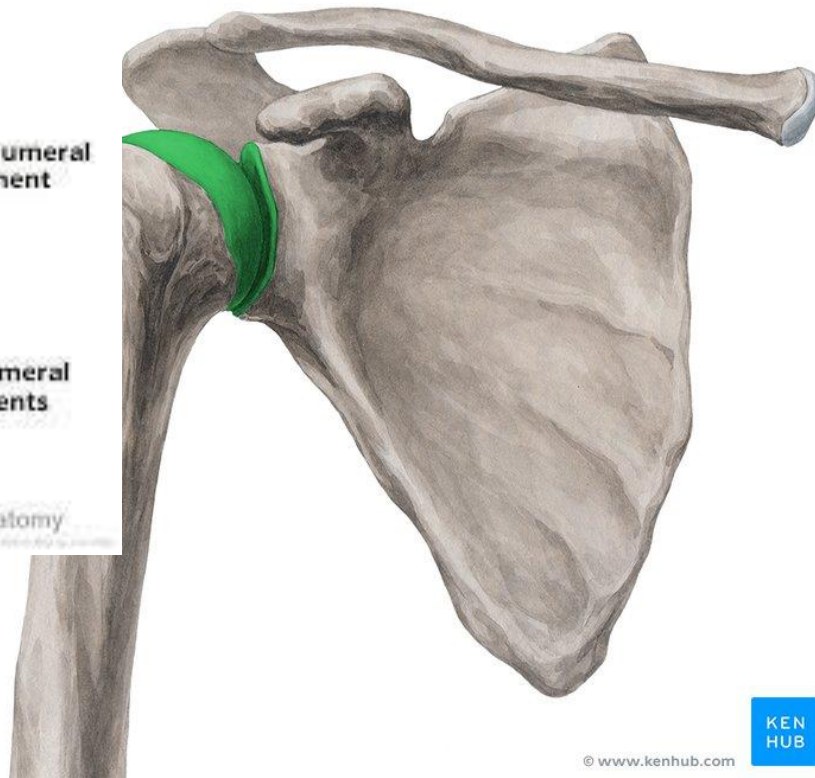
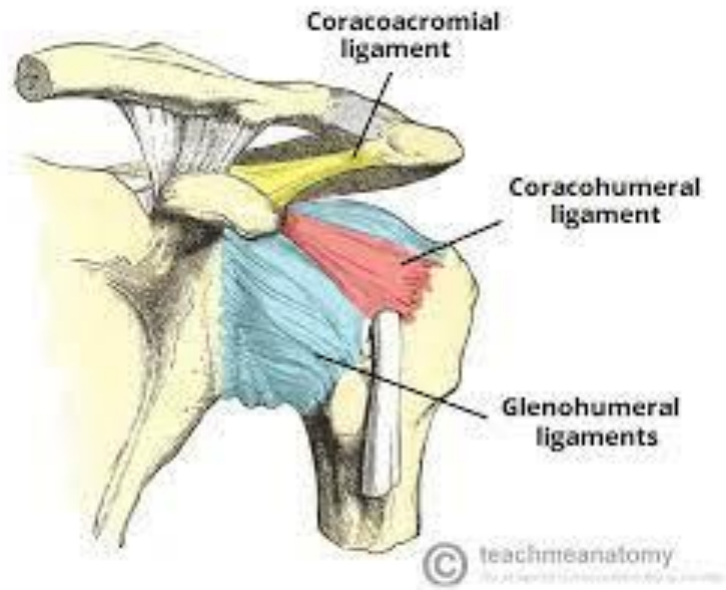
Prevention of sport injuries

- ▶ Warm Up
- ▶ • Cool Down
- ▶ • Hydration
- ▶ • Proper Technique
- ▶ • Equipment
- ▶ • Physical Conditioning
- ▶ • Facilities Management

Shoulder sport injuries

- ▶ Sternoclavicular injury
- ▶ • Acromioclavicular injury
- ▶ • Biceps tendon injury
- ▶ • • Superior labral injury
- ▶ • • Shoulder instability

Shoulder joint



Shoulder instability

- ▶ i.e head of humerus is not stable in glenoid
- ▶ The shoulder by virtue of its anatomy and biomechanics is one of **most unstable** and **frequently dislocated** joints in body
 - Its account nearly 50% of all dislocations



Dislocation



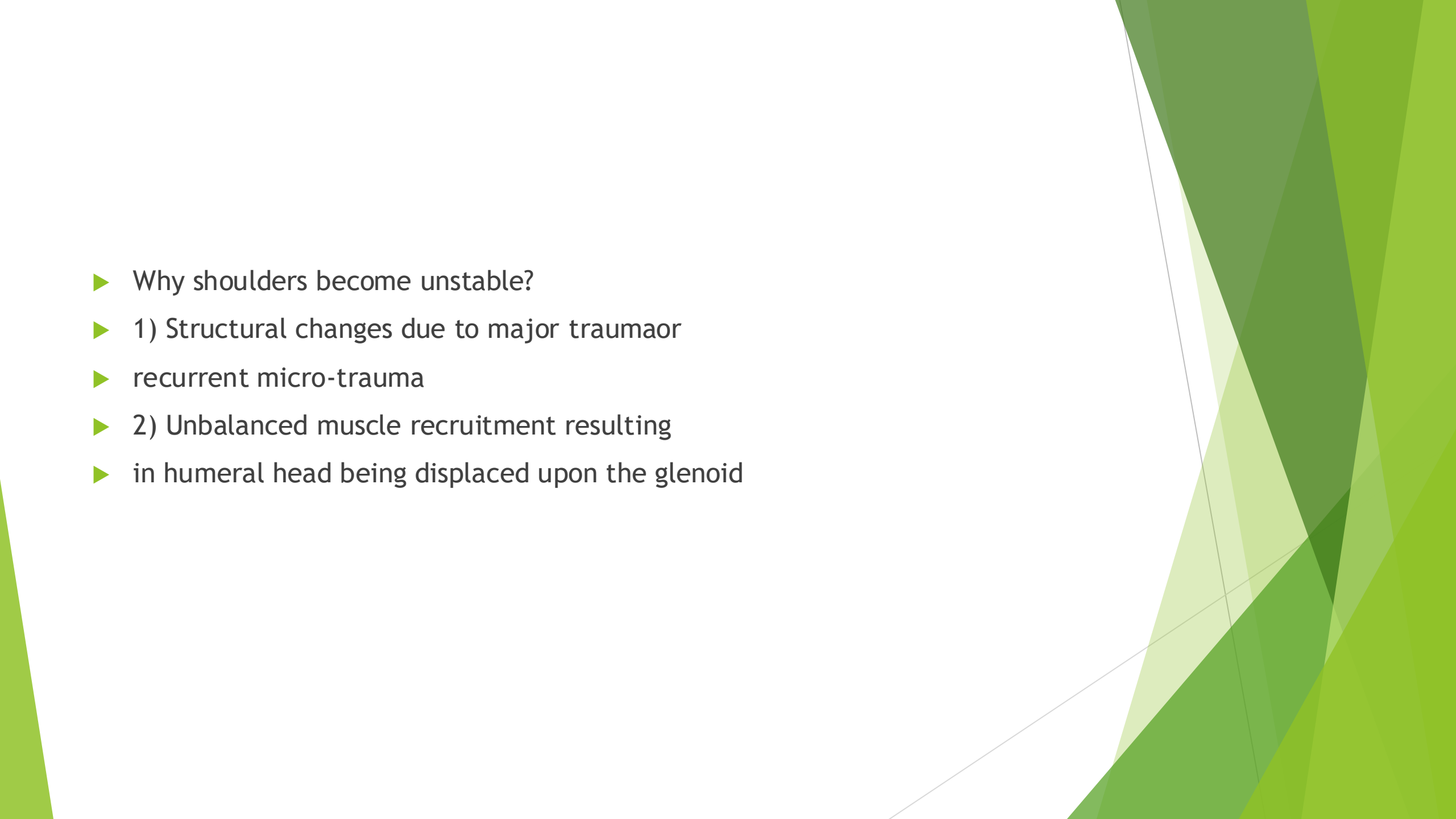
Normal
anatomy



Anterior
dislocation



Posterior
dislocation

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- The background of the slide features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern and dynamic visual effect.
- ▶ Why shoulders become unstable?
 - ▶ 1) Structural changes due to major trauma or recurrent micro-trauma
 - ▶ 2) Unbalanced muscle recruitment resulting in humeral head being displaced upon the glenoid

TYPES



- **Traumatic structural instability**
- **Atraumatic (or minimally traumatic) structural instability**

TRAUMATIC ANTERIOR INSTABILITY

- ▶ • Commonest type of instability (>95% cases)
- ▶ • Recurrent dislocation may results in:
 - ▶ • Bankart's lesion
 - ▶ • Hill-Sachs lesion
- ▶ • Recurrent subluxation alternately occur with
 - ▶ Dislocation
- ▶ • Patient age >50, often associated with tears of
 - ▶ rotator cuff



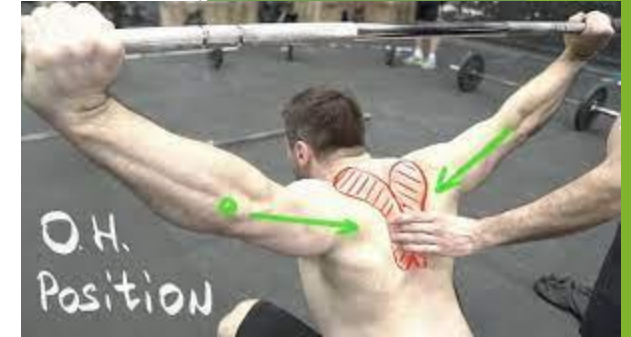
Hill sach



Clinical features:

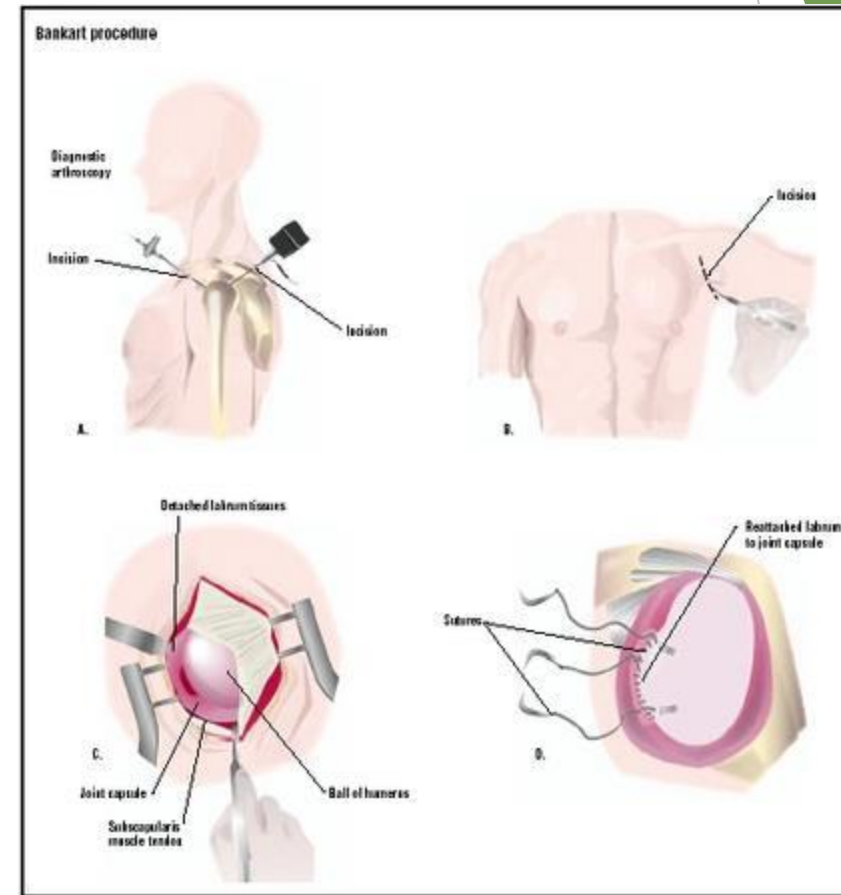
- ▶ • History of the shoulder “coming out”, perhaps during a sporting event
- ▶ • The 1st episode of acute dislocation: patient may be able to describe the mechanism precisely (ie: an applied force with the shoulder in abduction, external rotation and extension

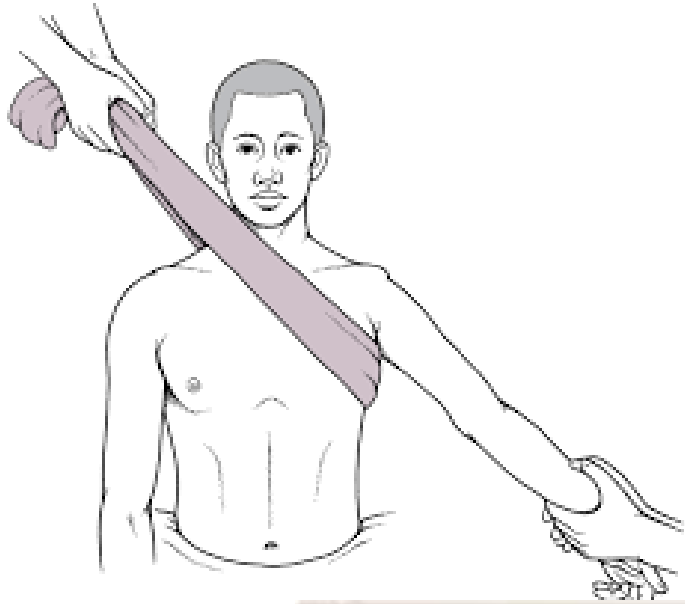
- ▶ Recurrent subluxation: symptoms and signs are less
- ▶ obvious
- ▶ A catching sensation □ numbness/weakness
- ▶ (“dead arm syndrome”)
- ▶ Whenever shoulder is used with the arm in
- ▶ the overhead position



treatment

- Relocation under anesthesia or sedation
- Dunlop, Kocher, Cunningham,,,,etc
- Surgical repair ...bankart repair
- Arthroscopic or open





Simon maneuver

Patient prone on table with affected limb hanging freely over edge; 10–15-lb weight suspended from wrist. Gradual traction overcomes muscle spasm and in most cases achieves reduction in 20–25 minutes.



Sport knee injuries

- ▶ one of the most commonly injured
- ▶ joints
- ▶ lack of bony and muscular support
- ▶ positioned between the 2 longest
- ▶ bones
- ▶ weight bearing and locomotion functions

Pelvic and hip injuries

- ▶ A stress fracture is a very small crack in the bone. This can happen from repetitive trauma and is commonly seen in athletes - particularly long-distance runners. Stress injuries can be found in the shin bone, foot, heel, hip and lower back. Treatment for stress fractures usually involves rest while the bone heals and changing your activity level to prevent another injury.

Extrinsic factors happen outside of the body

- ▶ Practicing incorrect training or sport technique.
- ▶ Having too rapid of a training program
- ▶ Changing the surface you exercise on,
- ▶ Using poor equipment or improper footwear (
- ▶ Doing repetitive activity in certain high-impact sports, such as:
- ▶ Long-distance running (tibia, hip).

Intrinsic factors

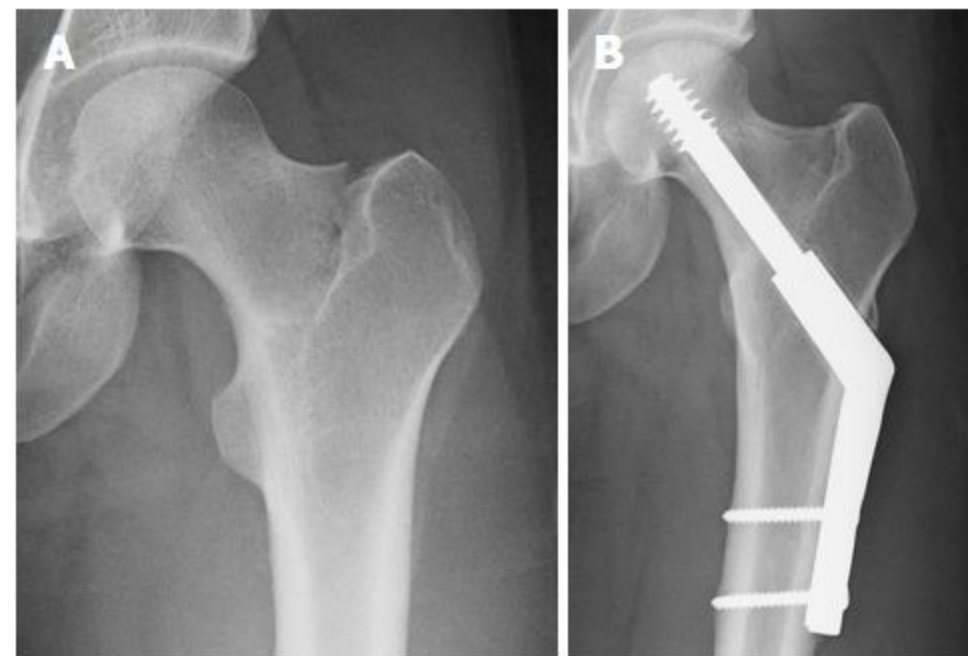
- ▶ are things that are related to the athlete or patient and aren't impacted by outside forces. These factors can include:
- ▶ Age: Older athletes may have underlying bone density issues such as osteoporosis. Already weakened bone will develop a stress reaction and/or fracture sooner than healthy bone.
- ▶ Weight: Both ends of the spectrum seem to be at risk for stress injuries. Someone with a low BMI or underweight individual may have weakened bones and someone with a high BMI doing repetitive loading with their body weight would also be at risk for injuries.
- ▶ Anatomy: Foot problems can affect the way the foot strikes the ground. These foot problems can include bunions, blisters, tendonitis, and low or high arches. Muscle weakness, imbalances or lack of flexibility can also be a factor.
- ▶ Sex: Females may be at risk if they have irregular menstrual periods or no periods.
- ▶ Medical conditions: Osteoporosis or other diseases that weaken bone strength and density (thickness). The weak or soft bones may not be able to handle the changes in activity

The symptoms of a stress fracture can include

- ▶ Pain, swelling or aching at the site of fracture.
- ▶ Pain that begins after starting an activity and then resolves with rest.
- ▶ Pain that's present throughout the activity and does not go away after the activity has ended.
- ▶ Pain which occurs while at rest, during normal activity or with everyday walking.
- ▶ Pain which is worse with hopping on one leg or an inability to shift weight/hop on affected leg/foot.

Dx

- ▶ Plain x ray ...normal in early stages,periosteal reaction ,fine crack ,displacement
- ▶ MRI
- ▶ Bone scan



treatment

- ▶ 1.conservative rest ,ice creep bandage walking aids
- ▶ 2.operative.. Femoral neck displacement,internal fixation

ACUTE KNEE INJURIES

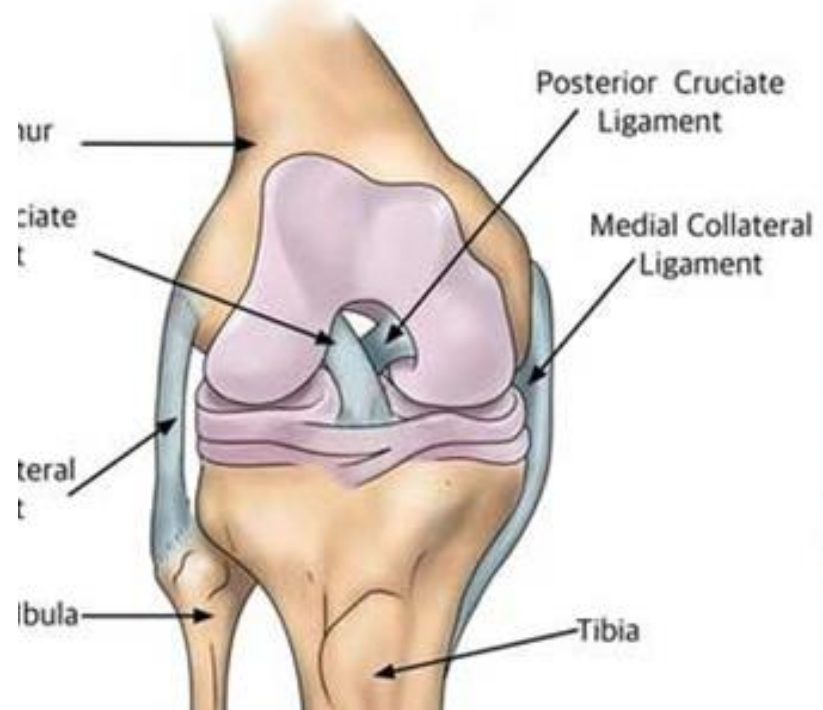
- ▶ 1. Anterior cruciate ligament (ACL) injury
- ▶ 2. Posterior cruciate ligament (PCL)injury
- ▶ 3. Medial Collateral ligament (MCL) Injury
- ▶ 4. Lateral collateral ligament (LCL) injury
- ▶ 5. Meniscal injuries
- ▶ 6. OSTEOCHONDRAL PROBLEMS
- ▶ 7. Patellar dislocation/instability

Anterior cruciate ligament (ACL)

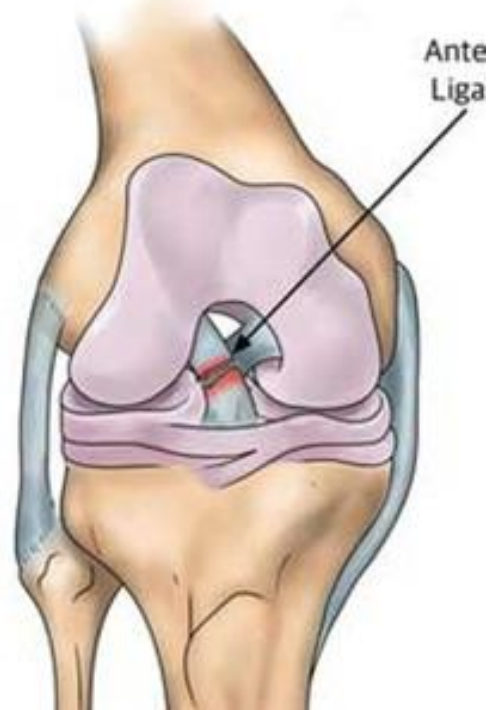
- ▶ □Most are non-contact injury, 2° to deceleration forces or hyperextension
- ▶ □Planted foot & sharply
- ▶ rotating



Normal ACL



Torn ACL



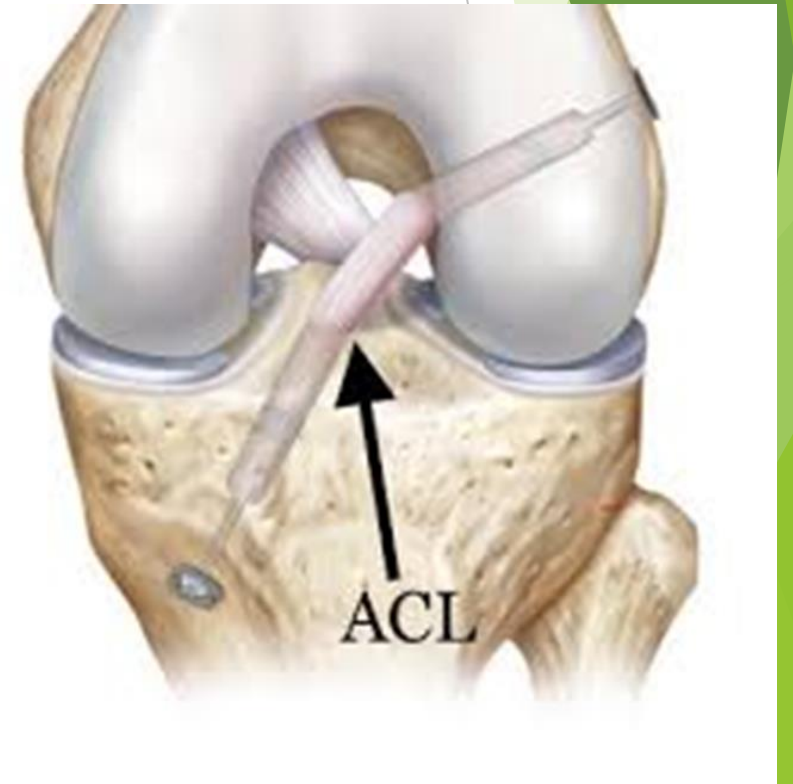
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- ▶ Hx:
 - ▶ □Hearing or feeling a “pop” & knee gives way
 - ▶ □Significant swelling quickly (< 1 hour)
 - ▶ □Unstable
 - ▶ □↓ range of motion (ROM)
 - ▶ □Achy, sharp pain with movement

MRI



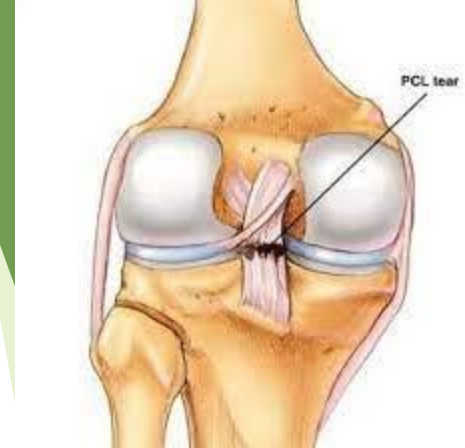
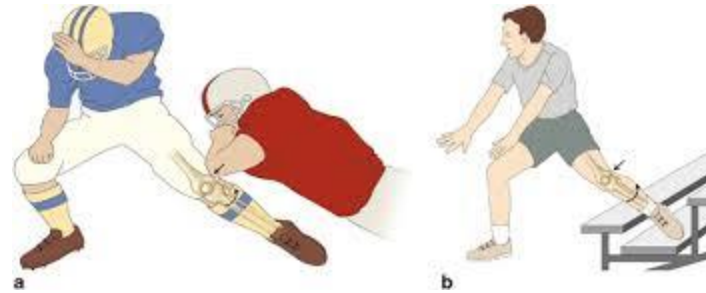
treatment

- ▶ Conservative treatment ...sedentary life elderly patients
- ▶ Operative ...ACL reconstruction young active and professional athletes



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PCL injuries



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Result from more severe trauma

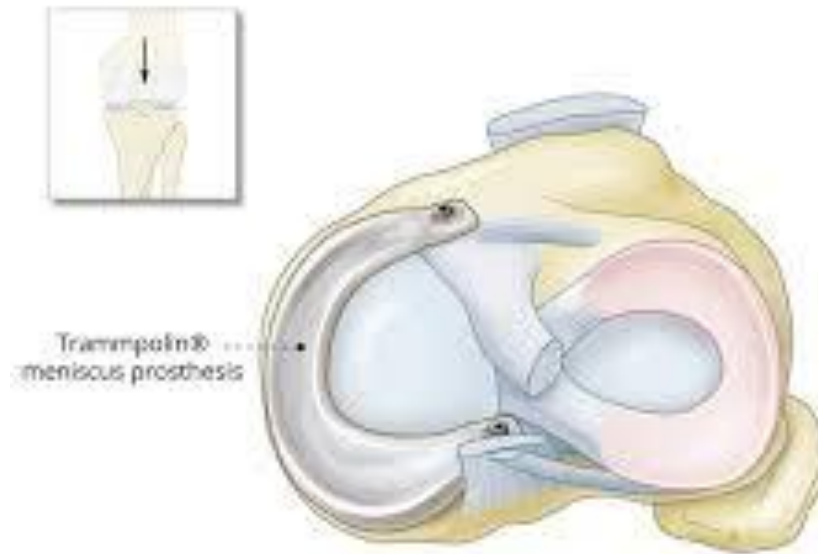
Associated with other ligaments like ACL LCL knee dislocation and intraarticular fracture

Dashboard injuries, motor cycles, and heavy sports



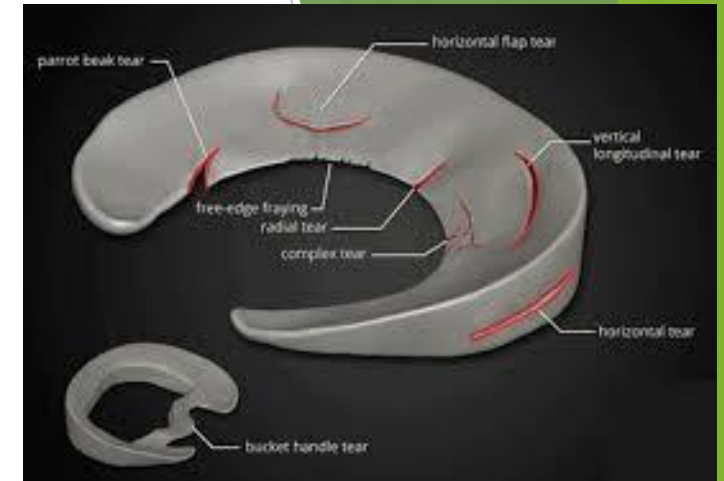
Meniscal injuries

- ▶ Meniscus = 'little moon' in greek
- ▶ Absorbs shock,
- ▶ distributes load,
- ▶ stabilizes joint
- ▶ Thick at periphery →
- ▶ thin centrally



Causes:

- ▶ Sudden twisting
- ▶ Young athletes
- ▶ Simple movements
- ▶ Older knee



Hx:

- ▶ Clicking, catching or locking
- ▶ Worse with activity
- ▶ Tends to be sharp pain at joint
- ▶ line
- ▶ Effusion

Treatment:

- ▶ RICE
 - ▶ Crutches
 - ▶ NSAIDs
 - ▶ Knee sleeve
- Surgical repair or
- ▶ excision (arthroscopic)
 - ▶ Asymptomatic tears do
 - ▶ not require treatment

Ankle sport injuries

- ▶ Foot and Ankle Sprains
- ▶ Foot and Ankle Fractures
- ▶ Tendon Ruptures
- ▶ Toe Fractures/Dislocations

Ankle sprain

- ▶ The most common injury in sports according to many sources
- ▶ • Accounts for 40% of all athletic injuries and 53% of basketball injuries
- ▶ 29% of soccer injuries
- ▶ • Peak incidence ages 15-19
- ▶ • Most resume normal activity within 4-8 weeks
- ▶ however 20-40% of those with severe sprains experience residual pain/instability

Grading Sprains

- ▶ Grade 1 - Stretch
 - ▶ » RTA: 1-2 weeks
- ▶ Grade 2 - Partial Tear
 - ▶ » RTA: 2-4 weeks
- ▶ Grade 3 - Complete Tear
 - ▶ » RTA: 8-10 weeks

Inversion Ankle Sprains

- ▶ Most common - accounts for 75% of all
- ▶ sprains
- ▶ Classic “rolled ankle”
- ▶ Structures of Interest
 - ▶ » Anterior Talofibular Ligament (ATFL)
 - ▶ » Calcaneofibular Ligament (CFL)
 - ▶ » Posterior Talofibular Ligament



Treatment of ankle sprains

► RICE



Medial Tibial Stress Syndrome

- ▶ • “Shin Splints”
- ▶ • Overuse injury - most often seen in runners
- ▶ • Often due to poor biomechanical support and stretching habits
- ▶ • Cause: Abnormal tension of muscle origin (TA/PT)
- ▶ • Although not seen as serious, can be quite disabling
- ▶ • Women 1.5 -3.5 times higher risk for this developing into stress fractures.



Achilles Tendon

Achilles Tendon Ruptures

- 1/4 are initially misdiagnosed as ankle sprains
- More common in men than women.
- Ages 30-40
- “Weekend Warrior”
- Other Risk Factors
- Fluoroquinolone treatment
- Cortisone injections



Mechanism

- ▶ • Sudden force plantarflexion
- ▶ • Violent dorsiflexion in a plantarflexed foot
- ▶ • Pathoanatomy:
- ▶ • Rupture can occur in 3 areas
- ▶ 1) Myotendinous Junction
- ▶ 2) “Watershed Area” (4-6 cm above calcaneal insertion) **
- ▶ 3) From Calcaneal Insertion



• Treatment

- ▶ Non-operative
 - ▶ • Casting/Functional Bracing in resting equinus
 - ▶ • Functional outcomes equal to that of operative management
 - ▶ • Increased Risk of Re-rupture
 - ▶ • Fewer complications
- ▶ Operative
 - ▶ • Open end to end repair

Classified as ..

- ▶ • Sports injuries can generally classified in one of two ways :
- ▶ Acute
- ▶ chronic

General Principles of Arthroscopy

- ▶ WHAT IS ARTHROSCOPY?
- ▶ This word arthroscopy came from GREEK ,
- ▶ "arthro" (joint) and "skopein" (to look).
- ▶ The term literally means to look within the joint
- ▶ simply as if you see a room through a key - hole
- ▶ instead of opening doors.
- ▶ It offers a high degree of accuracy combined with low morbidity for making diagnosis and offering treatment.



ARTHROSCOPE-

- It is a rigid optical instrument.
- Optical characteristics of an arthroscope are determined by diameter, angle of inclination and field of view.



INDICATIONS OF ARTHROSCOPY

► **DIAGNOSTIC**

- -For preoperative evaluation & confirmation of clinical diagnosis
- -For documentation in medicolegal cases

► ***THERAPEUTIC***

- •Smoothing of Torn cartilage
- •Damaged ligaments reconstruction
- •Loose bodies removal
- •Joint effusions
- •Biopsy procedures
- •Fracture fixation
- •Sports Related Injuries



ADVANTAGES OF ARTHROSCOPY

- ▶ Reduced postoperative morbidity
- ▶ Smaller incision
- ▶ Less intense inflammatory response
- ▶ Improved thoroughness of diagnosis
- ▶ Absence of secondary effects
- ▶ Neuromas, scars

- 
- ▶ Reduced hospital cost
 - ▶ Reduced complication rate
 - ▶ Improved follow-up evaluation : second-look
 - ▶ Possibility of performing surgical procedures that are
 - ▶ difficult to perform through open arthrotomy

COMPLICATIONS

- ▶ ● Damage to intraarticular structures: most common
- Damage to Menisci and Fat Pad
- Damage to Cruciate Ligaments
- Damage to Extraarticular structures
- Hemarthrosis
- Thrombophlebitis
- Infection
- Tourniquet Paresis
- Synovial Herniation and Fistulas
- Instrument Breakage

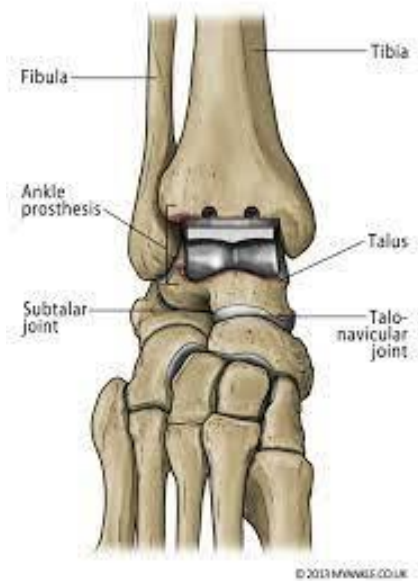
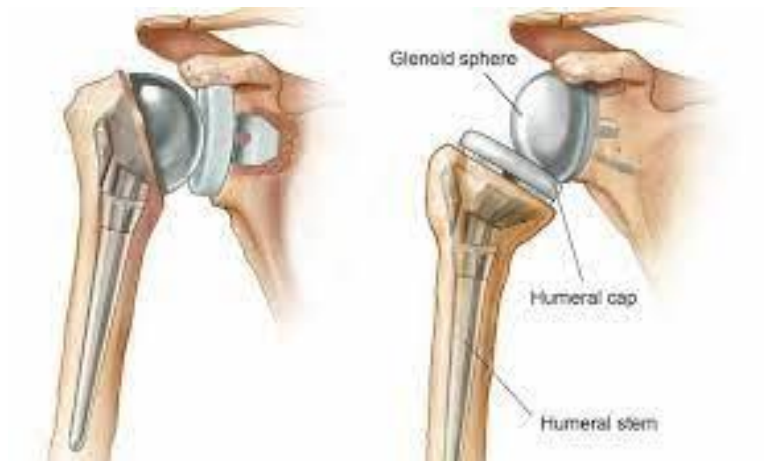
Arthroplasty

- ▶ Definition
- ▶ It is a replacement of the diseased joint with artificial joint
- ▶ Indications “painful joint caused by”
 - ▶ 1. advanced OA
 - ▶ 2. advanced rheumatoid arthritis
 - ▶ 3. Posttraumatic arthritis
 - ▶ 4. Avascular necrosis
- ▶ When these conditions failed to respond to conservative R



contraindication

- ▶ Paralytic disorders, poliomyelitis muscular dystrophies
- ▶ Charcot joints
- ▶ Infected joints
- ▶ Pediatric population



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Questions ??